Tiago Ferrão-Lindqvist

Test & Validation Engineer | Python, SQL & Data Automation

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PROFILE

Mechanical engineer with 12 years of experience in product design and test and validation of automotive and industrial components. Uses Python and MATLAB to process test-rig data, automate validation and reporting, and support engineering decisions. Experienced querying SQL Server test-results data for trend analysis. Interested in moving into data engineering.

EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components *Mar 2019–Present*
Seat structures and seat-adjuster mechanisms | Setúbal | Mar 2019–Present

- Plan and conduct durability, vibration and climatic tests across six servo-hydraulic and electric test rigs; prepare test reports for OEM customers.
- Developed Python tools using pandas, NumPy and Matplotlib to process rig output files, check every channel against test specifications and generate standard report figures. Reduced report preparation time from about three days to half a day per test.
- Query the plant's SQL Server test-results database to analyse trends across three product generations.
- Correlate rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis of field failures. Proposed a redesigned adjuster rail bracket that reduced warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive used to select rig load profiles.

Mechanical Design Engineer | Halvik Pumps *Sep 2014–Feb 2019*
Industrial centrifugal pumps | Porto | Sep 2014–Feb 2019

- Designed pump casings and impellers in SolidWorks and produced production drawings with GD&T; owned DFMEA for two product lines.
- Conducted structural FEA of casings in ANSYS. Supported a change from grey cast iron to ductile iron for one casing family, saving 12% in material cost.
- Supported conversion of the pump performance test bench to a PLC-controlled setup using Siemens S7.

EDUCATION

MSc Mechanical Engineering | Sado Valley Polytechnic 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

SKILLS

Python (pandas, NumPy, Matplotlib), SQL (SQL Server), MATLAB/Simulink, ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, Minitab, design of experiments (DOE), GD&T, DFMEA, 8D problem solving, Excel/VBA, Power BI

LANGUAGES

Portuguese (native), English (fluent), Spanish (good)